REMOTE SSH ACCESS TO WSL UBUNTU FROM ANDROID TERMUX

Project Report

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1. Objective:

To configure and access a WSL2 Ubuntu system remotely from an Android device using SSH and to troubleshoot network connectivity issues using practical networking and security concepts

2. Problem Statement:

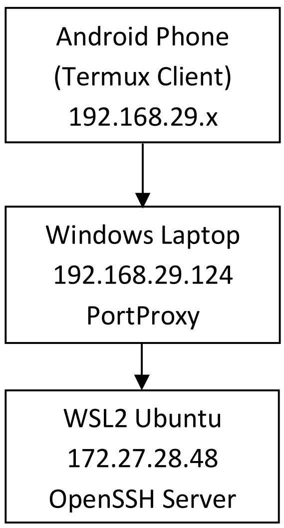
The objective was to remotely access a WSL2 Ubuntu system running on a Windows laptop from an Android device using the SSH protocol through the Termux application. During the implementation, multiple connectivity issues were encountered, including connection timeout errors, inability to reach the WSL virtual IP address from the mobile device, and challenges related to Windows firewall and network configuration. The project focused on identifying the root cause of these issues and implementing a reliable solution for successful SSH communication

3. System Environment:

Component	Specification
Client Device	Android phone with Termux
Client Application	Termux Terminal Emulator
Host Operating System	Windows 10 / Windows 11
Subsystem Environment	WSL2 (Ubuntu Linux)
Network Protocol & Port	SSH (Secure Shell) / Port 22

4. Network Architecture:

The project consists of an Android device running Termux, a Windows laptop hosting a WSL2 Ubuntu instance, and a Wi-Fi network connecting both devices. SSH communication is established from the Android device to the Windows host, which forwards the connection to the WSL2 Ubuntu instance through Windows PortProxy.



Architecture Description :

1 .Android phone acts as the SSH client.

2. Windows Laptop acts as the host system and network gateway

3. PortProxy forwards incoming SSH traffic from Windows to WSL2

4. WSL2 Ubuntu runs the OpenSSH server

5. Wi-Fi Network provides communication between the Android device and Windows host.

5. Commands Used:

Command	Purpose
ip a	Display network interfaces and ip addresses
ip route show	Display routing table
ping 192.168.29.124	Test connectivity between phone and laptop
ssh tejasrinu@IP	Establish SSH connection
sudo service ssh status	Check SSH service status
sudo ss –tlnp | grep :22	Verify SSH is listening on port22
ipconfig	Display Windows network configuration
netstat –an | findstr :22	Check listening ports on Windows
netsh interface portproxy show all	Verify port forwarding configuration

6. Troubleshooting Process:

Step by step :

Step 1: Verify WSL IP Address
Command: ip a
Observation: The WSL Ubuntu instance was assigned the IP address 172.27.28.48

Step 2: Verify SSH Service Status
Command: sudo service ssh status
Obsrvation: The SSH service was running successfully

Step 3: Verify SSH Listening port

Command: sudo ss –tlnp | grep :22
Observation: SSH was listening on 0.0.0.0:22, indicating that the service was accepting connections on all interfaces.

Step 4: Test SSH from Windows
Command: ssh tejasrinu@172.27.28.48 (Execute in Windows poweshell)
Observation: SSH login was successful from the Windows host system

Step 5: Test connectivity from Android
Command: ping –c 4 192.168.29.124
Observation: The Android device successfully reached the Windows laptop with 0% packet loss

Step 6: Analyze Windows Listening Ports
Command: netstat –an | findstr :22
Observation: Port 22 was listening only on 127.0.0.1, preventing external devices from accessing the SSH service.

Step 7: Identify Root Cause
Observation: The WSL IP address belonged to a virtual NAT network and could not be accessed directly from devices connected Wi-Fi.

Step 8: Configure Port Forwarding
Command: netsh interface portproxy add v4tov4 listenaddress=0.0.0.0 listenport=22 connectaddress=172.27.28.48 connectport=22
Observation: Incoming SSH traffic was forwarded from Windows to the WSL Ubuntu instance.

Step 9: Final Verification
Command: ssh tejasrinu@192.168.29.124
Result: The SSH connection was established successfully from Andriod Termux to WSL Ubuntu.

7. Root Cause Analysis:

During troubleshooting, it was observed that the Android device and Windows laptop were connected to the same Wi-Fi network and could communicate successfully through ICMP ping requests. However, SSH connections from the Android device consistently resulted in timeout errors.

Further investigation revealed that the WSL2 Ubuntu instance was assigned the IP address **172.27.28.48**, which belongs to a virtual NAT (Network Address Translation) network created by WSL2. This virtual network is isolated from the physical Wi-Fi network and is not directly reachable by external devices such as the Android phone.

Although the SSH service was running correctly inside WSL2 and could be accessed locally from the Windows host, external devices could not directly communicate with the WSL2 IP address. As a result, SSH requests sent from the Android device never reached the OpenSSH server running inside WSL2.

Therefore, the root cause of the problem was the network isolation created by the WSL2 virtual NAT environment, which prevented direct access to the WSL2 IP address from devices on the local Wi-Fi network.

Issue	Analysis
SSH Timeout	Android device could not reach WSL2 directly
WSL2 IP	Located on a virtual NAT network
SSH Service	Running correctly
Phone-to-PC Connecting	Working successfully
Actual Cause	WSL2 network isolation and lack of port forwarding

8. Solution:

After identifying that the WSL2 Ubuntu instance was isolated behind a virtual NAT network, a solution was implemented using Windows PortProxy and Windows Firewall configuration. Port forwarding was configured to redirect incoming SSH connections from the Windows host to the WSL2 Ubuntu instance. Additionally, firewall rules were updated to allow inbound TCP traffic on Port 22. This enabled external devices on the local Wi-Fi network to access the SSH service running inside WSL2.

9. Results:

The project was successfully completed by establishing secure SSH connectivity between an Android device running Termux and a WSL2 Ubuntu instance hosted on a Windows laptop. The initial connection timeout issues were identified, analyzed, and resolved through systematic troubleshooting and network configuration.

Results Achieved

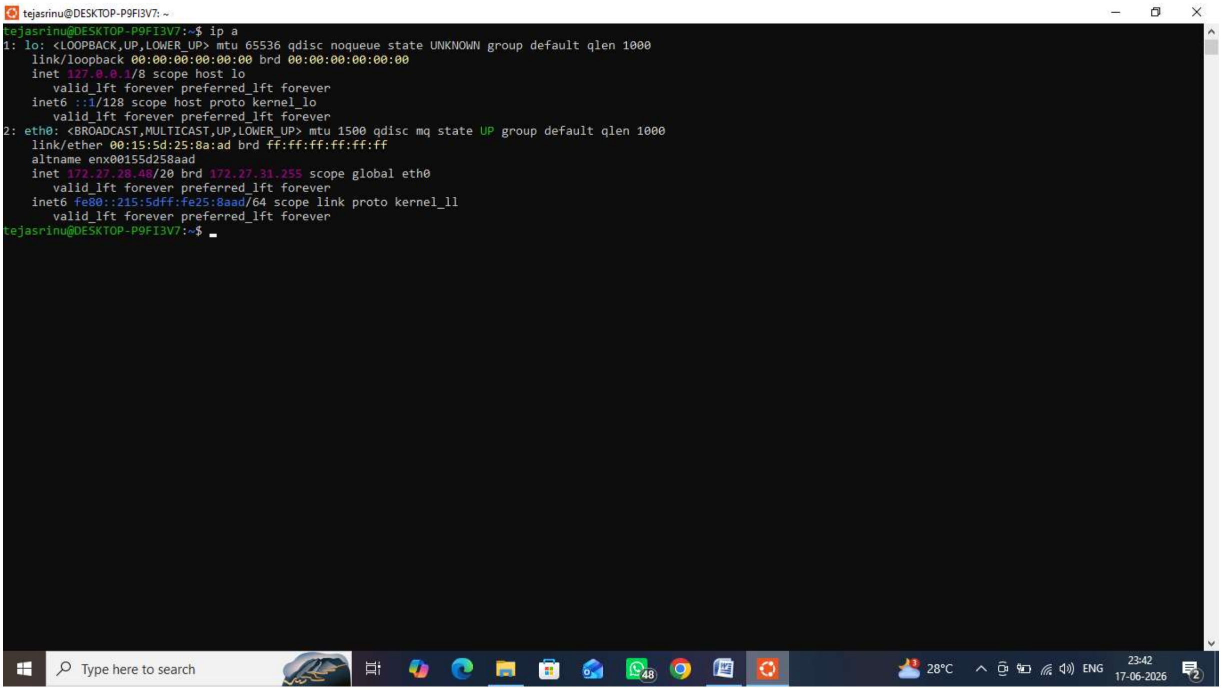
1. Successfully connected Android Termux to WSL2 Ubuntu using SSH.

2. Verified proper operation of the OpenSSH server.
3. Implemented Windows PortProxy for SSH traffic forwarding.
4. Configured Windows Firewall to allow inbound SSH connections.
5. Confirmed network communication between Android and Windows systems.
6. Resolved connectivity issues caused by WSL2 NAT network isolation.

10. Screenshots:

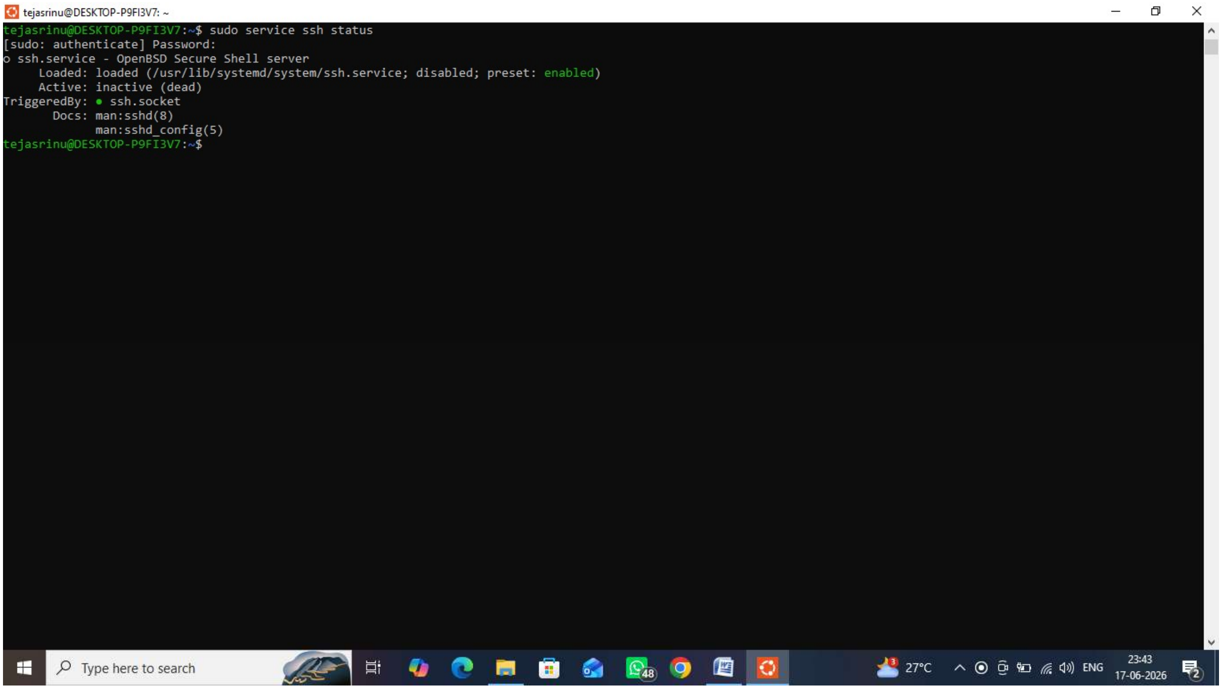
Screenshot 1 – WSL IP Adress

WSL2 Ubuntu IP Address Verification using ip a



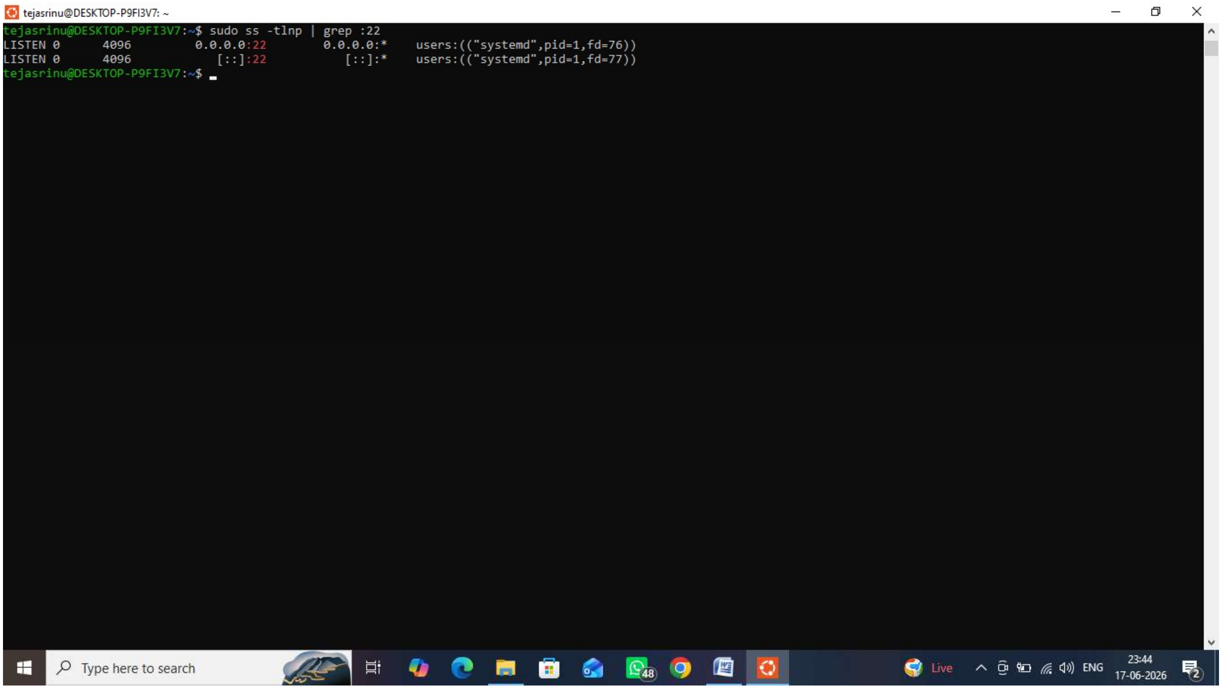
Screenshot 2 – SSH Service Status

OpenSSH Service Running Sucessfully



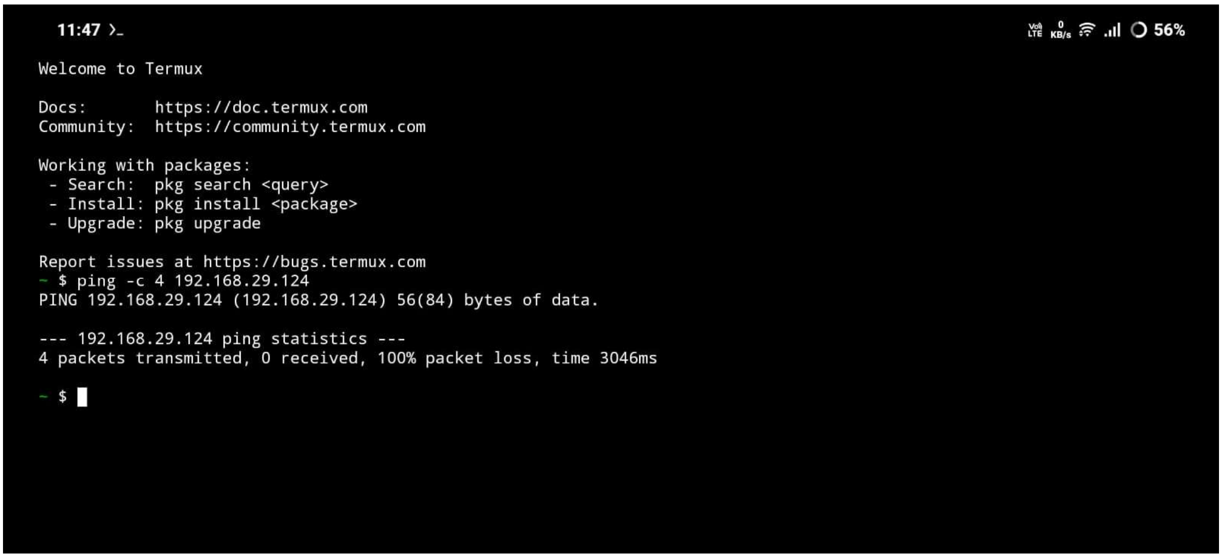
Screenshot 3 – SSH Listening on Port 22

Verification of SSH Listening Port



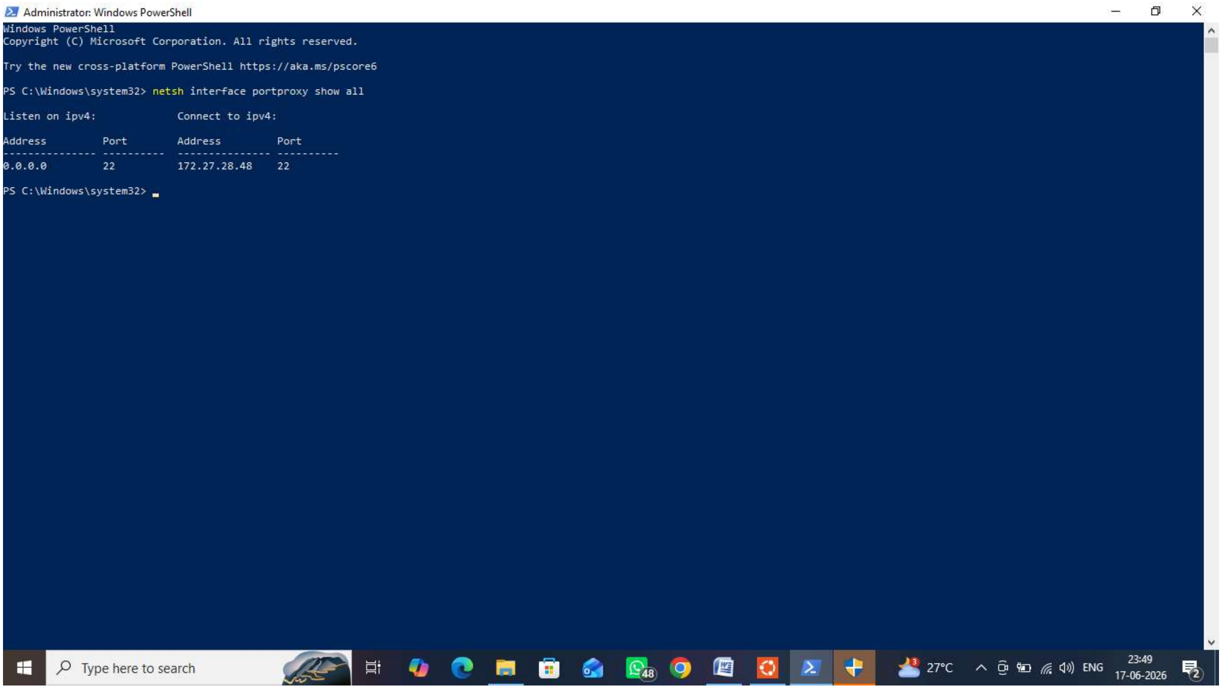
Screenshot 4 – Network Connectivity Test

Successful Ping Test from Android Device



Screenshot 5 – PortProxy Configuration

Windows PortProxy Configuration



Screenshot 6 – Successful SSH Login

Successful SSH Connection from Android Termux to WSL2 Ubuntu

```
~ $ ssh tejasrinu@192.168.29.124
tejasrinu@192.168.29.124's password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 6.18.33.1-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Wed Jun 17 18:20:12 UTC 2026

  System load:  0.17                Processes:             31
  Usage of /:   0.2% of 1006.85GB   Users logged in:       0
  Memory usage: 10%                 IPv4 address for eth0: 172.27.28.48
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Tue Jun 16 16:39:45 2026 from 172.27.16.1
tejasrinu@DESKTOP-P9FI3V7:~$
```

11. Conclusion:

This project successfully established SSH connectivity between an Android device running Termux and a WSL2 Ubuntu environment hosted on a Windows laptop. The SSH timeout issue was analyzed and resolved using Windows PortProxy and firewall configuration. Through this project, practical knowledge of SSH, TCP/IP networking, NAT, port forwarding, and network troubleshooting was gained. The project achieved its objective and demonstrated real-world problem-solving and system administration skills.